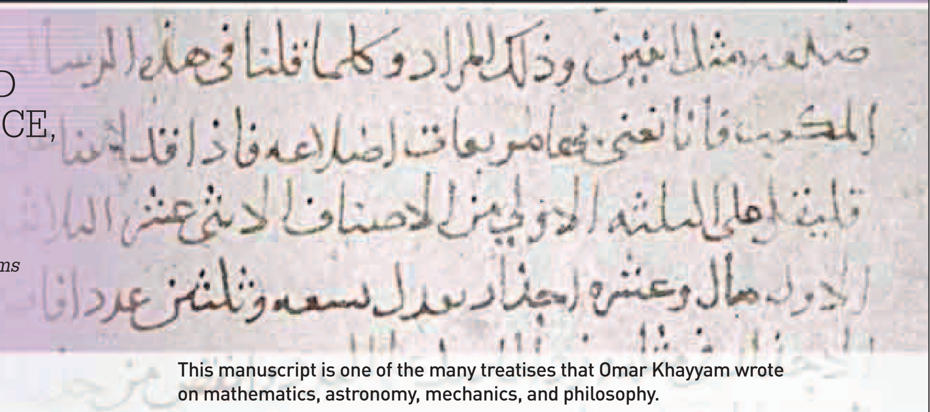
> **“** BY THE **HELP OF GOD** AND WITH HIS PRECIOUS ASSISTANCE, I SAY THAT **ALGEBRA** IS A **SCIENTIFIC ART. ”**

Omar Khayyam, from *Treatise on Demonstration of Problems of Algebra*, 1070



This manuscript is one of the many treatises that Omar Khayyam wrote on mathematics, astronomy, mechanics, and philosophy.

PERSIAN MATHEMATICIAN AND ASTRONOMER Omar Khayyam began work on his ***Treatise on Demonstration of Problems of Algebra*** in 1070, the year that he moved to Samarkand, Uzbekistan, and devoted himself to study and writing. In it, he gave a complete **classification of the types of cubic equation** (an equation involving a term to the power of three, such as $x+y^3=15$) and described for the first time a general theory for solving them using geometry. The method he used involved the use of conic sections and curves. He realized that some equations, such as quadratic (involving a squared term) and cubic equations, had more than one solution.

Khayyam was also an accomplished astronomer: from 1073 he worked at the observatory in Isfahan, Iran. Much of his work was concerned with the compilation of **astronomical tables**, but he also helped improve the accuracy of the calendar.

Bamboo
Shen's discovery of fossilized bamboo in a cool, dry area led him to conclude that the region would have been warm and humid in the past.

At Isfahan, he also worked on his poetry, later collected in ***The Rubaiyat of Omar Khayyam***. In 1079, he calculated the **length of a year** as 365.24219858156 days—a greater degree of precision than ever before, and remarkably close to the modern measurement of 365.242190 days. This led to the introduction of a **new calendar in the Islamic world**, which was more accurate than the Julian calendar used in Europe at the time.

Meanwhile, in China, the polymath **Shen Kuo** retired from a successful career as a civil servant and military leader in the court of the Song dynasty, and devoted his time to study. He wrote an extraordinarily wide-ranging collection of essays on subjects as diverse as politics, divination, music, and the sciences. The *Dream Pool Essays*, named after his garden estate, were finished in 1088, and included an overview of the sciences of the time, as well as some innovative ideas. For example, Shen was the first to give a description of the **magnetic compass needle**. He explained how it could be used in navigation to determine the direction of North. He also contributed to the fields of paleontology and geology. Describing the discovery of the remains of **marine creatures** in the strata of a cliff hundreds of miles from the coast, he suggested that these must have been covered by silt over a long period of time—which would have been later eroded—and also proposed that the cliff must have at some time been a coastal area. He described **fossilized bamboo** unearthed by a landslide, in an area where bamboo does not grow, and came to the conclusion that this was the remains of an ancient forest from a time when the climate of the area had been significantly different.

OMAR KHAYYAM (1048–1131)

Born in Persia (now Iran), Omar Khayyam showed a talent for astronomy and mathematics at an early age; he wrote many of his treatises before he was 25 years old. In 1073, he was invited by Sultan Malik-Shah to set up an observatory in Isfahan. Here, he worked on calendar reform and astronomical tables, before returning to his home town.

> **“** UNDER THE GROUND, A **FOREST OF BAMBOO SHOOTS** WAS REVEALED… THESE WERE SEVERAL DOZENS OF FEET **BELOW THE PRESENT SURFACE** OF THE **GROUND. ”**

Shen Kuo, from *Dream Pool Essays*, 1088

1070 Omar Khayyam begins writing ***Treatise on Demonstration of Problems of Algebra***

1079 Omar Khayyam calculates the **length of a year** and enables **calendar reform**

1088 Shen Kuo completes ***Dream Pool Essays***